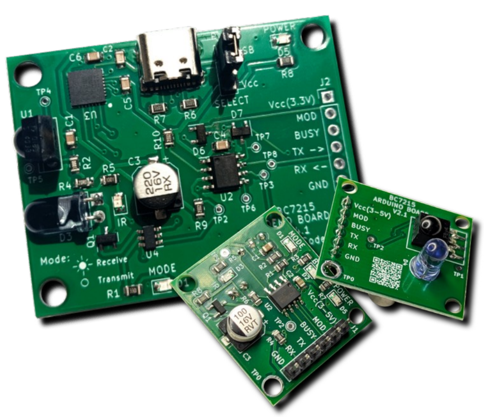
BC7215A

Development Board

&

Utility Software

User Manual



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Introduction

The BC7215A is a universal infrared (IR) remote control codec (transceiver) chip. It supports the majority of IR remote control formats and can also be used for IR data transmission using remote control data formats since it inputs and outputs raw data.
Two versions of the development board are available (USB and non-USB), both offering identical IR functionality. These boards allow for:

- Decoding data from any IR remote control.
- Duplicating any IR remote signal.
- Sending custom data using any IR remote format.
- Demonstrating IR data communication (requires two boards).
- Offline control of any AC and decoding IR AC commands.

Installation

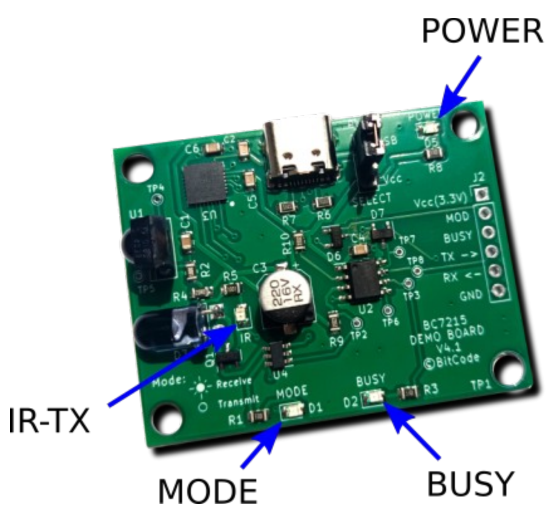
Hardware

USB Interface Development Board

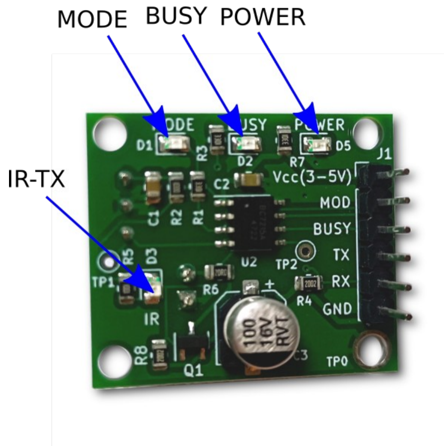
This board is plug-and-play with a computer via USB and requires no external power. It includes pin headers for power and control signals to connect to user hardware (similar to the Arduino/MCU board). A jumper selects between USB power or external power from pin headers.

LED Indicators:

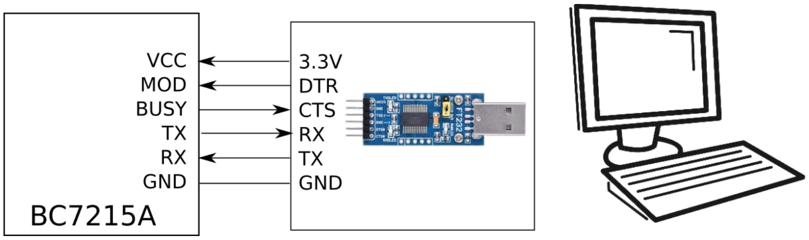
- POWER – Power indicator
- MODE – Receive/Transmit mode; lights up during Receive mode.
- BUSY – Serial busy indicator (hardware flow control).
- IR-TX – IR emission indicator; lights up during transmission.



Arduino/MCU board



This version lacks a USB interface and provides pins for connection to user circuits. To use it with a computer, a USB-to-UART converter with CTS and DTR signals is required for hardware flow control and mode switching.



Software

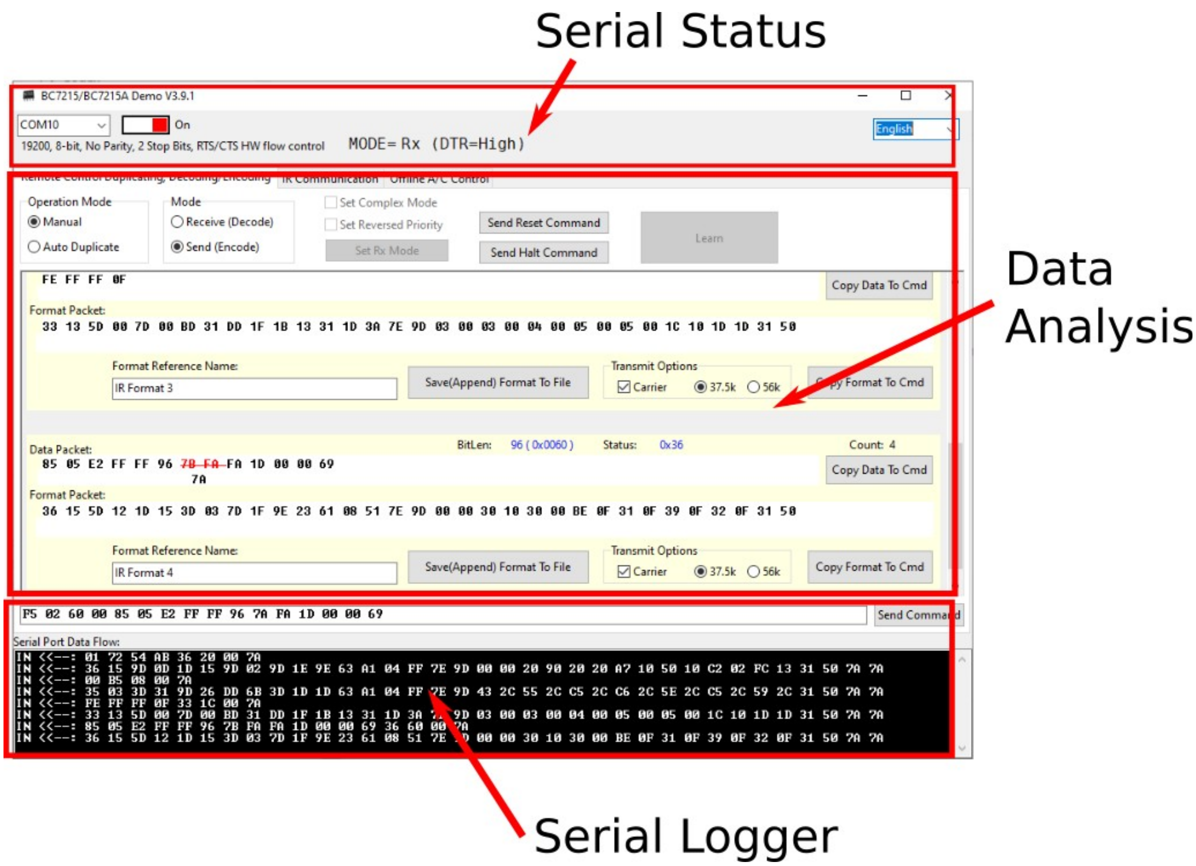
USB Driver

Windows typically installs the driver automatically. If manual installation is needed, download the driver corresponding to the USB chip (e.g., CP2102). Note: The default CH340 driver in Linux does not support hardware flow control; additional attention is required for Linux systems.

Demo(Utility) Software

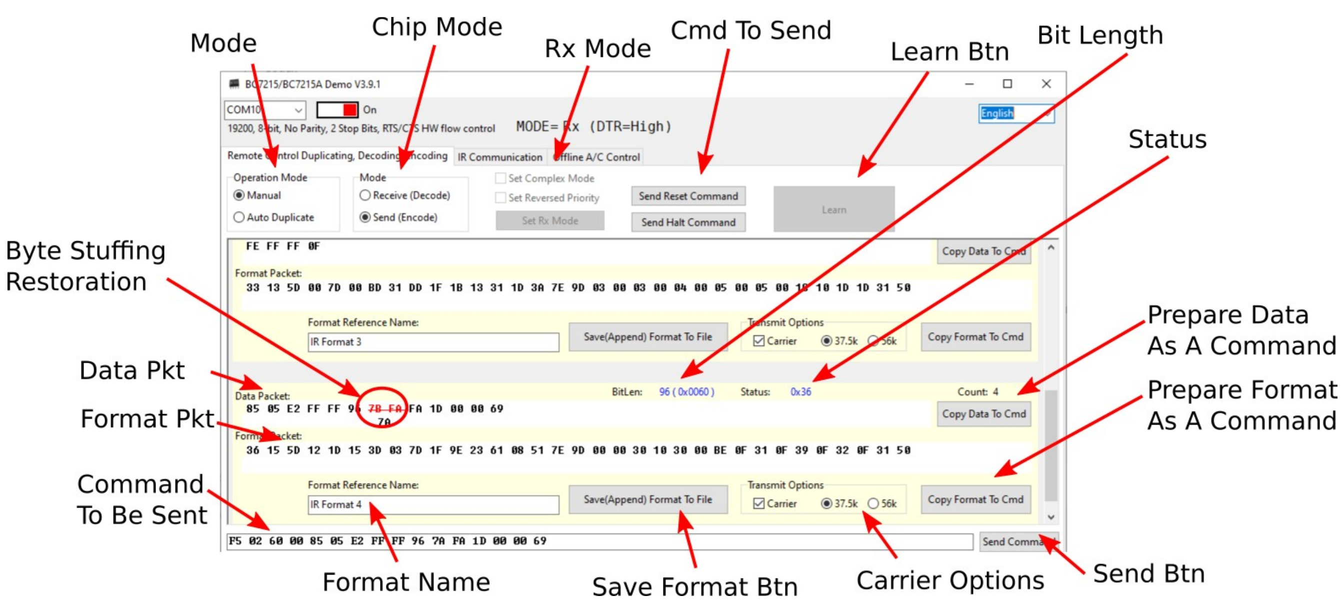
A Windows-based demo software is provided as a standalone EXE file that requires no installation. It consists of three areas:

1. Serial Status (Top): Shows connection and flow control status.
2. Data Analysis (Middle): Displays decoded results and settings.
3. Serial Data (Bottom): Shows raw hex data sent/received over the serial port.



Feature Details

Remote Control Codec Interface



Operation Mode

Includes "Manual" and "Auto Duplicate". In Auto Duplicate mode, clicking "Learn" puts the software in a waiting state; once a signal is received, it automatically switches to transmit mode and copies the data to the command area for re-transmission.

Chip Working Mode

Allows manual switching between Receive (Decode) and Transmit (Encode).

Set Mode

Choose between Simple (data packets only) or Complex (data + format packets) decoding modes.

Reset & Sleep Commands(In TX mode)

BC7215A can accept these 2 commands in TX mode, please refer to the BC7215A datasheets for details.

Learn Button

This button is only functional when the software is in "Auto Duplicate" mode. Clicking it puts the software into a receiving wait state. Once a valid signal is received, the data is automatically copied to the transmission area, the software enters Transmit mode, and the signal can be duplicated by clicking the Send Command button.

BitLen

This field displays the total number of bits in the received remote control signal.

Signature

This is the feature word of the received remote control signal. Detailed information can be found in the BC7215A datasheet.

Copy Data To Cmd

This button rearranges the raw data packet of the remote signal and adds the instruction header F502 before copying it to the command transmission section. If the chip is in Transmit mode, clicking the Send button will then issue the IR transmission command.

Copy Format to Cmd

This button copies the format information (if available) to the transmission area and adds the instruction header F601. If the BC7215A is in Simple mode, this button will be disabled as no format information is output.

Send Command

When the chip is in Transmit mode, clicking this button sends the data from the "Command to be Sent" area out through the serial port.

Transmit Options

The BC7215A contains an internal carrier generator. The user can control the carrier switch and frequency. These controls are included in the transmitted format information and are copied to the command area alongside the format packet.

Note: Use caution when disabling the carrier function. When disabled, a continuous high current will flow through the IR LED, which may cause damage.

Save Format to File

Users can save format information to a file. Using this button, format data can be saved as a new file or appended to an existing one.

Format Reference Name

Note that this name refers to the label the user gives to the saved format information, not the filename itself.

Command to be Sent

This is an edit box where data is placed after using the copy buttons. Data must be represented in 2-digit hexadecimal format. Users can edit this data directly but must ensure the correct format and byte count. For details on format download and IR emission commands, refer to the BC7215A datasheet.

Data here can be the raw data before byte stuffing (i.e., it can contain 0x7A and 0x7B). Byte stuffing will be applied automatically during serial transmission.

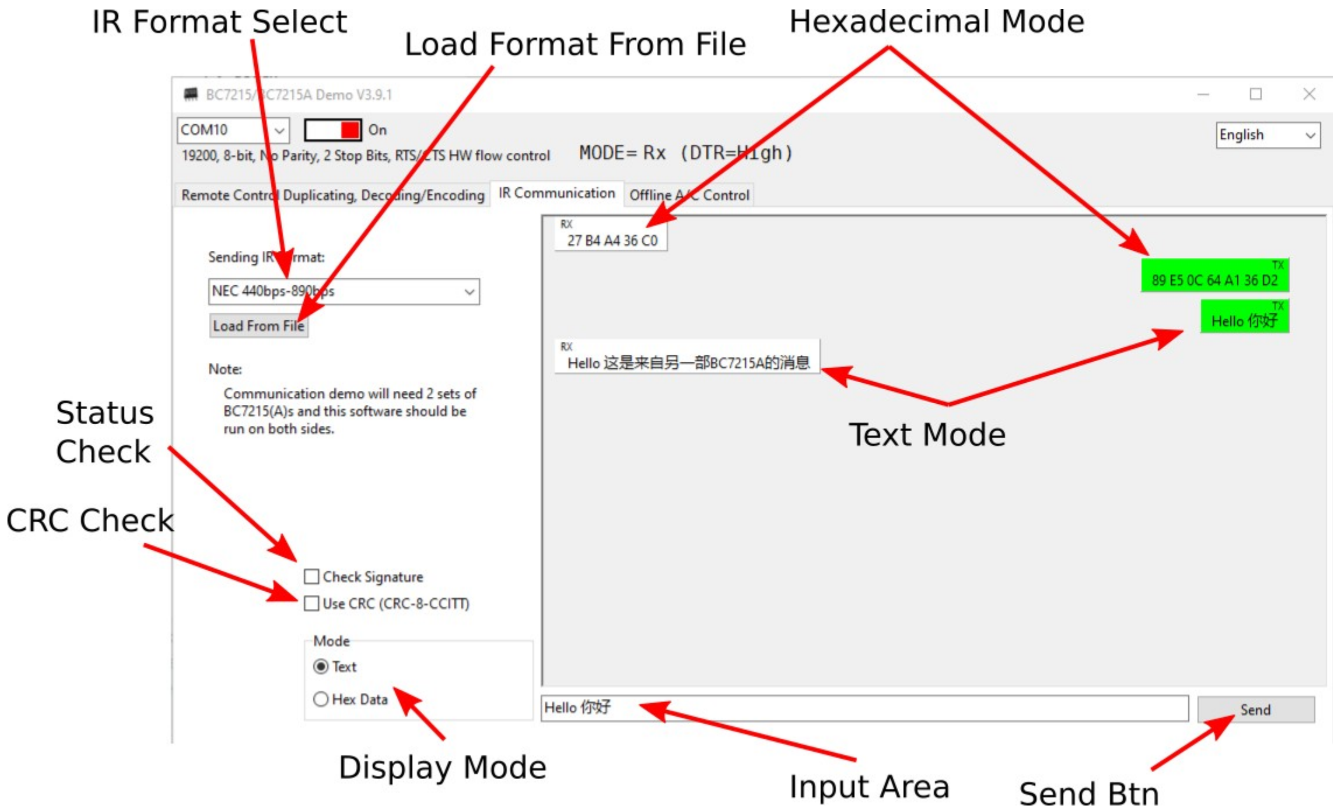
Data Packet & Format Packet

These areas display the raw data and format information, respectively.

Restoration of Byte Stuffing Encoding

Red text with a strikethrough in the raw data indicates a byte that was stuffed; its original value (0x7A or 0x7B) is displayed below it.

Communication Demo Interface



Sending IR Format

The software includes two built-in formats. Users can also use their own saved encoding formats.

Load From File

Allows users to load their own IR formats, typically those saved from the Codec interface.

Communication Content Window

Simulates a chat application, with options for Hex or Text display.

Mode (Display Mode Selection)

Selects whether the content is shown as Hex or Text. In Text mode, the input is converted using UTF-8 encoding before sending, and received data is interpreted as a UTF-8 string.

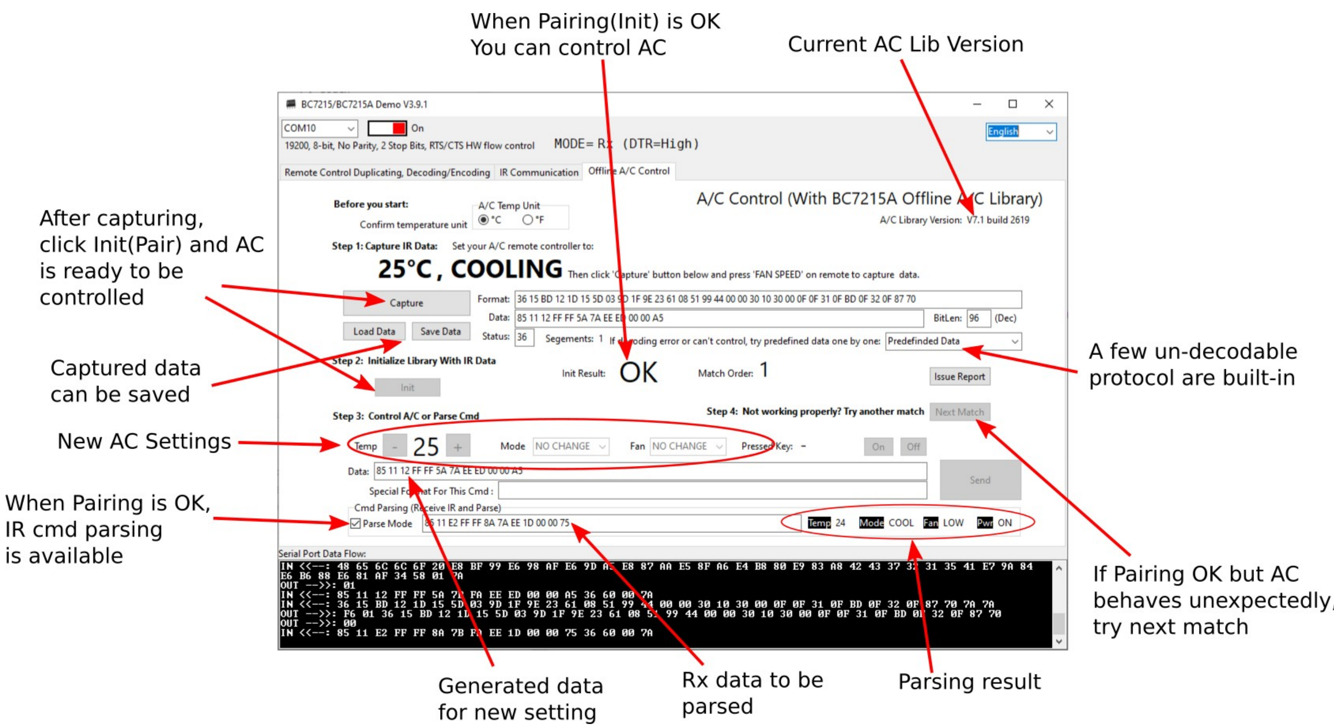
USE CRC

When enabled, the software appends a CRC byte to the end of sent data. For received data, the last byte is treated as the CRC. If the CRC check fails, the data is marked as a communication error.

Check Signature

Since the BC7215A is a universal receiver, it may pick up environmental IR noise. Users must verify if received data is from the intended source. Checking the signal's Signature is a fast screening method. If this option is enabled and the received Signature differs from the one used for transmission, the data is marked as an error.

Offline AC Control Page



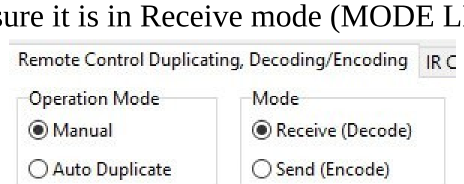
This section demonstrates the use of the BC7215A offline AC control library to control any air conditioner.

- Control can be achieved in three simple steps following the on-screen prompts.
- Predefined Data: For specific AC remote formats that are currently not supported by BC7215A, the AC library includes built-in formats and data. If capture fails, users can try these predefined datasets.
- Analysis Function: Starting from version V3.5, the software can parse temperature, mode, fan speed, and power status from the received IR signals after successful initialization.

Typical Experiments

1. Remote Control Decoding

(1) Connect the board and ensure it is in Receive mode (MODE LED on).



(2) Aim the remote at the IR receiver and transmit. A flashing BUSY LED indicates success, and results will appear in the software.

2. One-Key Signal Duplication

(1) Set the software to Auto Duplicate mode.

(2) Click the Learn button.

(3) Transmit from the remote; the software will automatically switch to Transmit mode upon receipt.

(4) Click **Send Command** to duplicate the signal.

3. Sending Custom Data via Remote Format

Method A (Via One-Key Duplication):

(1) Duplicate the signal as in Experiment 2.

(2) Manually modify the data in the "Command to be Sent" box (refer to BC7215A datasheet for format).

(3) Click **Send Command**.

Method B (Manual Mode):

(1) Set board to Receive mode.

(2) [Optional] Set to Complex mode and click **Set Mode**.

(3) Decode a remote signal.

(4) Switch to Transmit mode.

(5) [Optional] Configure carrier settings.

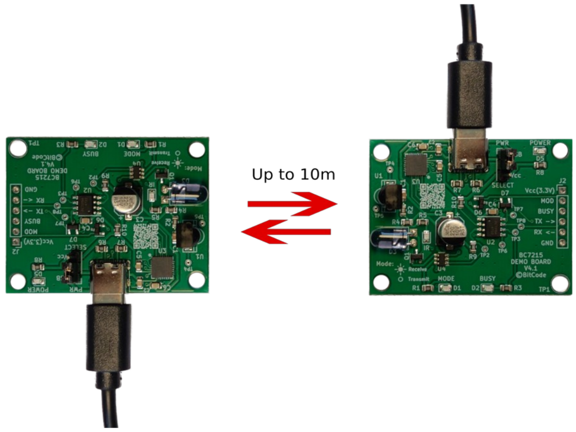
(6) [Optional] Use **Copy Format to Cmd** and click **Send Command** to download format to BC7215A.

(7) Click **Copy Data to Cmd**, manually modify the data, and click **Send Command**.

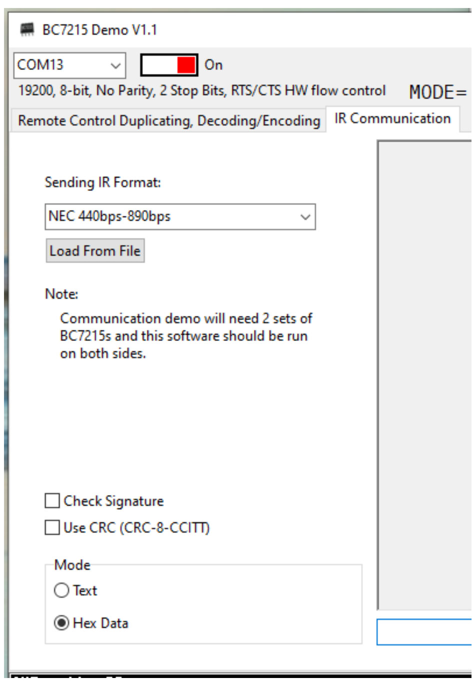
Note: If data length changes, the length field in the IR command must be updated. Also, specific formats (like NEC) may require data integrity (e.g., inverse code bytes); otherwise, the target device may not respond.

4. IR Data Communication (Requires 2 Boards)

Connect two boards to different computers or USB ports and run two instances of the software. Align the IR transmitters and receivers. The range is approximately 10 meters. Select Hex Data mode and choose the same IR format and CRC settings on both sides.



Input hex values (e.g., A7 35 2D 89) separated by spaces. It is recommended to use formats with Signature 0x34 for data communication.。



Note:

- The demo software has two built-in IR encoding formats: the NEC format and a format used for a certain type of air conditioner remote control, with a bit rate slightly higher than that of the NEC format. Since the BC7215A can receive signals of any format and cannot specify a receiving format, it is possible for the two sides to use different formats; however, it is recommended that both sides use the same format.
- The demo software allows the use of any encoding format supported by the BC7215A for infrared communication by loading format files (obtained by saving from the remote control codec interface). However, some infrared formats have special requirements for data. For

example, the RC5 format requires the two highest bits of the first byte of data to be 1, which makes it inconvenient to use for data communication; therefore, it is not recommended as an encoding format for infrared data communication. It is recommended to use a format with a Signature (feature word) of 0x34 for data communication.

- In actual use, because the BC7215A may receive infrared signals that are not sent from the data source, it is recommended to add special marker bytes to the data and use a certain verification mechanism, such as a CRC check, to reduce unnecessary interference to the receiving end.

5. IR Text Chat (Requires 2 Boards)

Settings are identical to Experiment 4, but the mode is set to Text. Due to the low IR bitrate (approx. 100-150 bytes/sec), avoid long messages. The BC7215A has a maximum limit of 512 bytes per transmission. Text is sent via UTF-8 encoding.

Caution, if you have different CRC settings on each side, it may cause CRC error or even make the text non-displayable.

Built-in IR Encoding Formats

NEC 440bps-890bps：

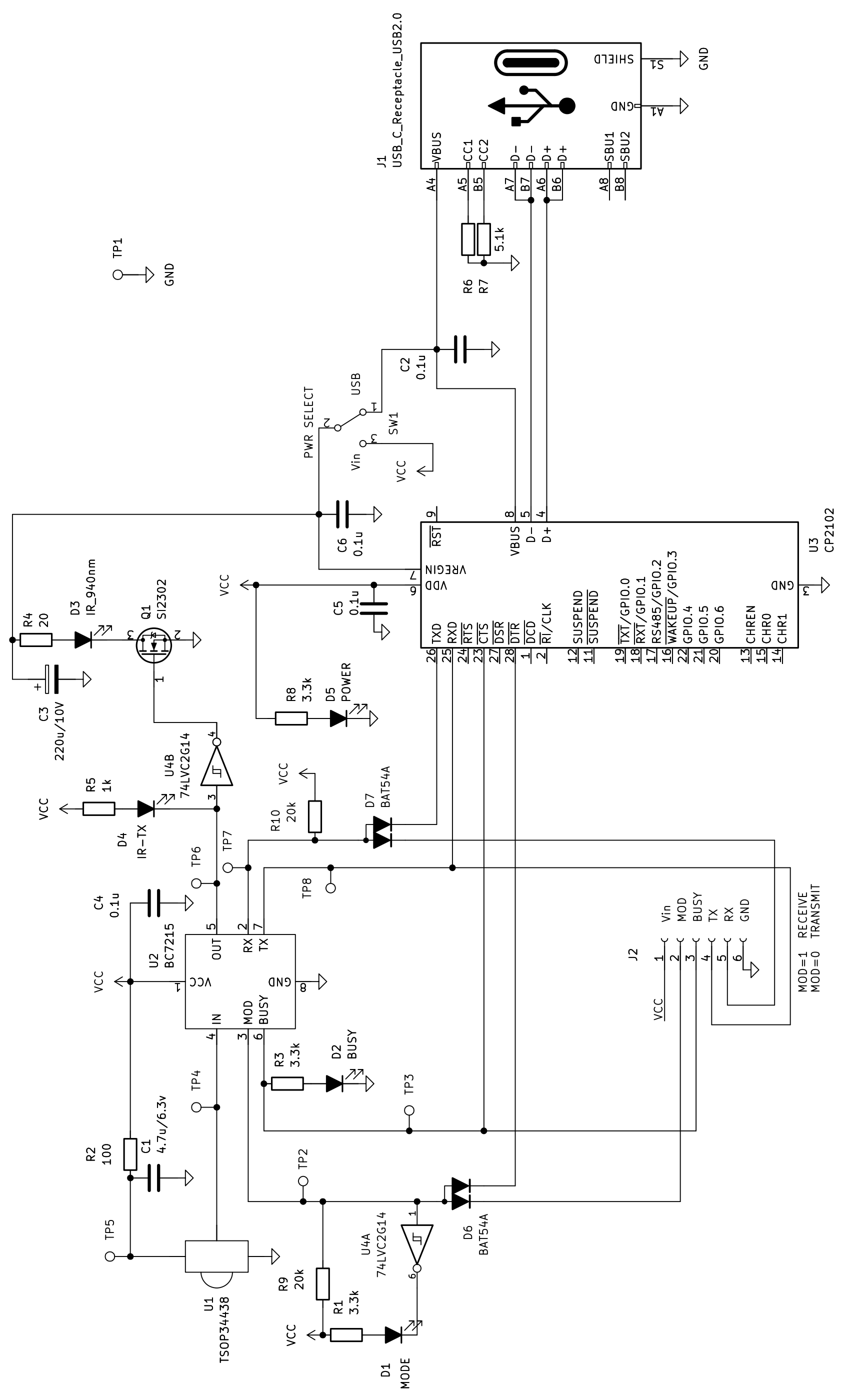
```
34 14 5D 0D 5D 14 3D 3D 1D 1C 9C 62 A0 29 B2 99 44 00 00 C2 36 9F F7
FA B8 E2 9A A3 26 EA 90 87 30
```

SANYO AC 610bps-1220bps：

```
34 1B 9D 16 FD 1B 9D 0A 5D 1C 9C 33 59 0C C8 07 0B 00 00 20 00 20 10
20 00 40 00 07 10 07 10 07 10
```

Schematics

USB Development Board



Arduino/MCU Board